

Flux Studio

Complete Project Overview

RunPod Serverless Case Study - FLUX.1-dev Image Generation

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Repository: <https://github.com/germainssafari/Runpod>

Model: FLUX.1-dev (Black Forest Labs)

Platform: RunPod Serverless

Live endpoint: flux-studio-v2 - ID 7n4p3czd3nphae

1. Executive Summary

Flux Studio is an end-to-end AI image generation platform. A user types a text prompt in a ChatGPT-style browser interface. The React frontend calls a FastAPI backend, which submits an async job to RunPod Serverless. A GPU worker loads FLUX.1-dev, generates a PNG, and returns base64 image data. The UI displays the result in a chat thread with history, library, projects, and pinning.

Three-tier architecture

Layer	Technology	Role
GPU worker	Python + Diffusers + Docker	Runs FLUX.1-dev on RunPod GPUs
API proxy	FastAPI (Render)	Secrets, job submit, status polling
Chat UI	React + Vite (Vercel)	ChatGPT-style prompt to image

2. What We Built

Deliverable	Status	Location
Serverless handler	Done	serverless/handler.py
Dockerfile	Done	serverless/Dockerfile
RunPod endpoint	Done	flux-studio-v2
Working inference	Done	Logs + UI screenshot
ChatGPT-style UI	Done	app/frontend/
API proxy	Done	app/backend/main.py
CLI test script	Done	scripts/test_endpoint.py
GitHub repository	Done	Public repo
Documentation	Done	docs/

3. System Architecture

Flow: Browser -> FastAPI (Render) -> RunPod Serverless -> GPU Worker -> base64 PNG -> Chat UI

```
React Chat UI (Vercel)
  | POST /api/chat/submit
  v
FastAPI Proxy (Render)
  | POST /v2/{endpoint_id}/run
  v
RunPod Serverless (flux-studio-v2)
  | handler(job)
  v
GPU Worker (Docker + FLUX.1-dev)
  | { image_base64, prompt, width, height }
  v
Chat UI displays generated image
```

Why three tiers: (1) RunPod API keys stay server-side on Render. (2) Long GPU jobs use async submit + polling instead of blocking HTTP. (3) CORS is controlled via FRONTEND_URL.

4. Repository Structure

```
Runpod/  
|-- serverless/           handler.py, Dockerfile (GPU worker)  
|-- app/backend/          FastAPI -> RunPod proxy  
|-- app/frontend/         React chat UI  
|-- scripts/              test_endpoint.py, PDF generators  
|-- docs/                 Documentation + screenshots  
|-- render.yaml           Render blueprint  
L-- start.ps1             Local startup script
```

5. Phase 1 - Serverless GPU Worker

The RunPod worker is a Python script invoked per job. It loads FLUX.1-dev once (cached), generates an image, and returns base64 PNG data.

Key handler design (serverless/handler.py)

- Sequential CPU offload - keeps GPU VRAM low on 24 GB cards
- VAE slicing/tiling - reduces peak memory during decode
- Version marker v3-sequential-offload - visible in RunPod logs

```
HANDLER_VERSION = "v3-sequential-offload"

pipeline = FluxPipeline.from_pretrained("black-forest-labs/FLUX.1-dev", ...)
pipeline.enable_sequential_cpu_offload()
pipeline.vae.enable_slicing()
pipeline.vae.enable_tiling()

return {"image_base64": "...", "prompt": "...", "width": 512, "height": 512}
```

Docker image (serverless/Dockerfile)

- Base: nvidia/cuda:12.4.1-cudnn-runtime-ubuntu22.04
- Pinned: torch 2.5.1 (cu124), diffusers 0.32.2, transformers 4.47.1
- Image: docker.io/sfrgermain/runpod-flux-studio:latest or :v3

6. Phase 2 - Docker Image on Docker Hub

After building locally, the worker image was pushed to Docker Hub. RunPod pulls this image when starting workers.

Repositories / runpod-flux-studio / General

sfrgermain/runpod-flux-studio

Last pushed about 12 hours ago · Repository size: 15.4 GB · 0 · 164

Add a description

Add a category

General Tags Image Management Collaborators Webhooks Settings

Tags DOCKER SCOUT INACTIVE [Activate](#)

This repository contains 2 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	about 12 hours
v3		Image	less than 1 day	about 12 hours

[See all](#)

Repository overview INCOMPLETE

An overview describes what your image does and how to run it. It displays in the [public view of your repository](#) once you have pushed some content.

[Add overview](#)

Docker commands [Public view](#)

To push a new tag to this repository:

```
docker push sfrgermain/runpod-flux-studio:tagname
```

buildcloud

Build with
Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

[Go to Docker Build Cloud](#) →

Figure: Docker Hub - sfrgermain/runpod-flux-studio (tags: latest, v3, ~15.4 GB)

7. Phase 3 - RunPod Serverless Endpoint

7.1 Endpoint overview

Endpoint name: flux-studio-v2 - ID: 7n4p3czd3nphae

API URL: <https://api.runpod.ai/v2/7n4p3czd3nphae/run>

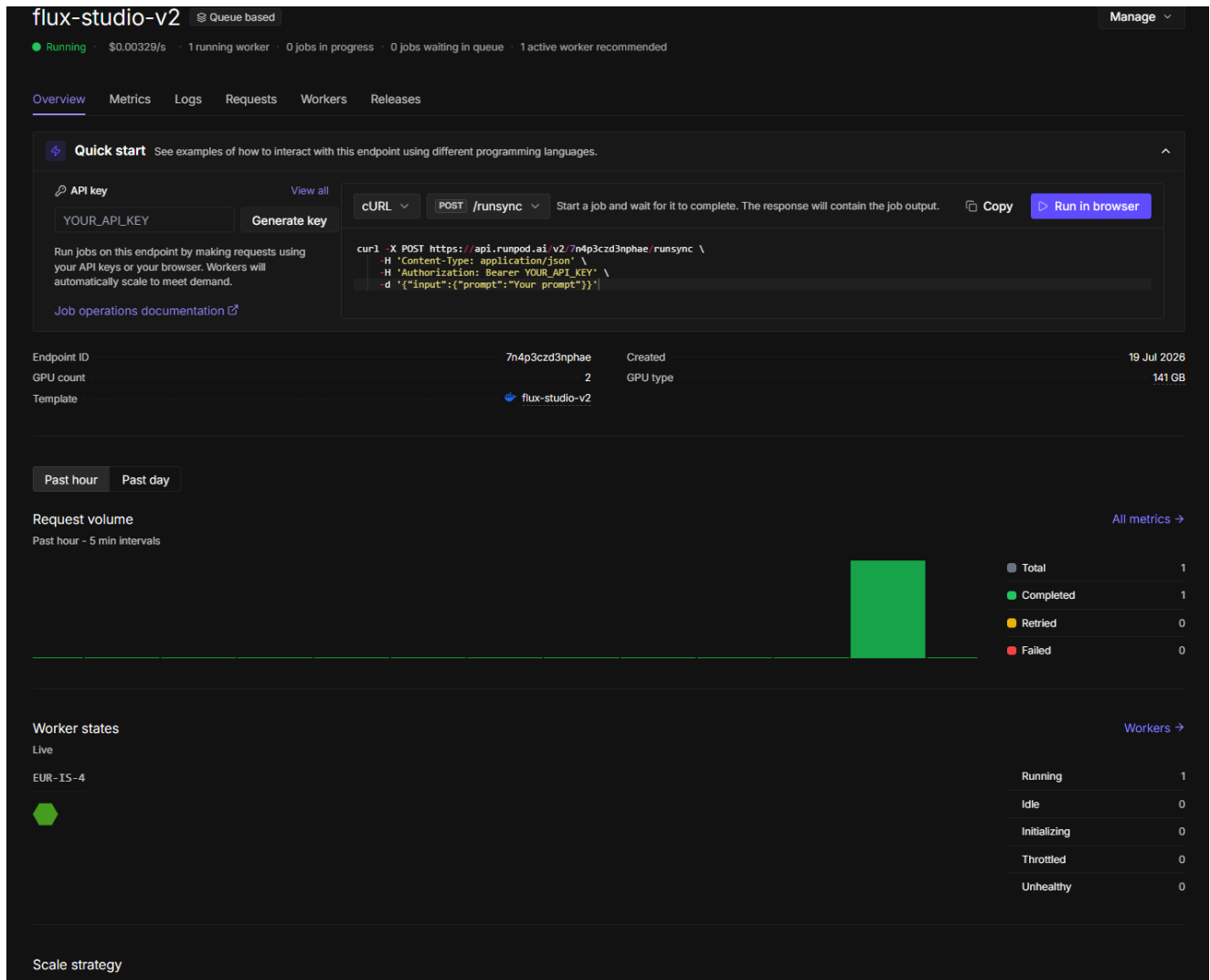


Figure: RunPod endpoint overview - flux-studio-v2, ID 7n4p3czd3nphae, Running

7.2 GPU and scaling configuration

- GPU: 141 GB VRAM tier (H200/H100 class)
- Max workers: 1 - Active workers: 1 (warm worker for demos)
- Idle timeout: 3600 s - Execution timeout: enabled

Edit endpoint

Endpoint name

flux-studio-v2

GPU configuration

☒	141 GB	1st	High Supply	\$0.00329/s
☐	16 GB		Low Supply	\$0.00032/s
☐	24 GB		High Supply	\$0.00038/s
☐	24 GB PRO		High Supply	\$0.00061/s
☐	32 GB PRO		High Supply	\$0.00088/s
☐	48 GB		High Supply	\$0.00068/s
☐	48 GB PRO		Medium Supply	\$0.00097/s
☐	80 GB PRO		High Supply	\$0.00253/s
☐	96 GB PRO		High Supply	\$0.00194/s
☐	180 GB PRO		Low Supply	\$0.00480/s

Max workers

The maximum number of workers that can be deployed at the same time.

Consider increasing max workers to 2 for better performance.

Active workers

Active workers always stay running to reduce response times.

0-1 active worker recommended based on current usage.

GPU count

Number of GPUs per worker.

Idle timeout

Set the duration a worker remains active (running) after its last request before scaling down. Workers still incur charges until they are marked idle.

☒ Enable execution timeout

Execution timeout

Figure: Endpoint configuration - 141 GB GPU, max/active workers = 1, idle timeout 3600s

7.3 Workers tab

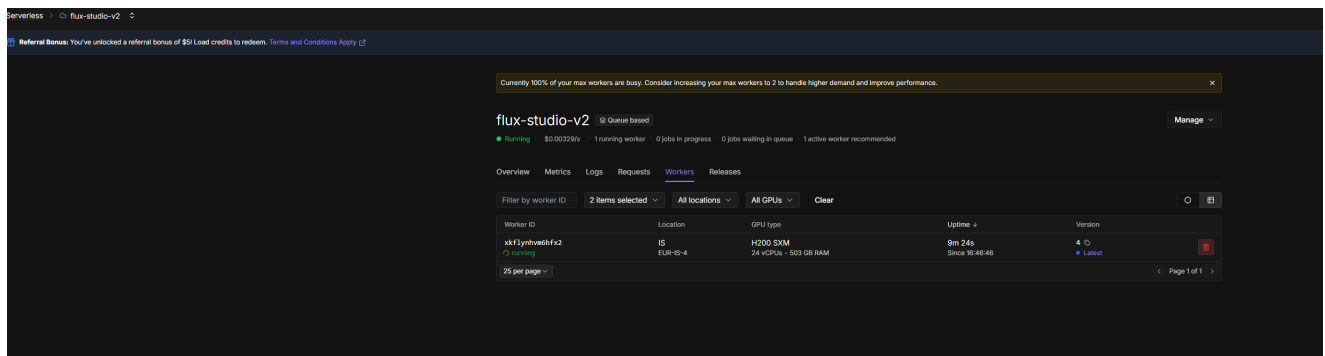


Figure: Workers tab - H200 SXM worker running (Version 4)

7.4 Worker telemetry



 \$6.69 



flux-studio-v2 / xkf1ynhvm6hfx2 

 Running |  H200 SXM |  IS

[Telemetry](#) | [Logs](#) | [Connect](#) | [Request History](#)

 Disk usage

34 GB (42%) 80 GB

 Volume usage

No volume

 Uptime

9m 51s

 # Processes

200

 CPU load

0%

INTEL(R) XEON(R) PLATINUM 8568Y+

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Figure: Worker telemetry - GPU, disk (80 GB), RAM, CUDA status

7.5 Logs - successful generation

Logs confirm fitness checks, model loading, and inference. Example prompt: 'A futuristic city with flying cars at night, neon reflections, cyberpunk' at 512x512.

flux-studio-v2 Queue based Manage

Running \$0.00329/s 1 running worker 0 jobs in progress 0 jobs waiting in queue 1 active worker recommended

Overview Metrics Logs Requests Workers Releases

Search logs Request ID Live All workers All levels Select date Ask AI

TIMESTAMP	LEVEL	WORKER	MESSAGE
Jul 19 16:47:20.659	INFO	xkflynhvm6hfx2	This container image and its contents are governed by the NVIDIA Deep Learning Container License.
Jul 19 16:47:20.659	INFO	xkflynhvm6hfx2	By pulling and using the container, you accept the terms and conditions of this license:
Jul 19 16:47:20.659	INFO	xkflynhvm6hfx2	https://developer.nvidia.com/ngc/nvidia-deep-learning-container-license
Jul 19 16:47:20.659	INFO	xkflynhvm6hfx2	A copy of this license is made available in this container at /NGC-DL-CONTAINER-LICENSE for your convenience.
Jul 19 16:47:20.659	INFO	xkflynhvm6hfx2	Starting Flux Studio worker [v3-sequential-offload]...
Jul 19 16:47:20.659	INFO	xkflynhvm6hfx2	--- Starting Serverless Worker Version 1.10.1 ---
Jul 19 16:47:20.659	INFO	xkflynhvm6hfx2	Running 7 fitness check(s)...
Jul 19 16:47:20.659	INFO	xkflynhvm6hfx2	GPU binary test passed: 2 GPU(s) healthy (CUDA 12.8)
Jul 19 16:47:20.659	INFO	xkflynhvm6hfx2	Memory check passed: 1918.496GB available (of 2015.37GB total)
Jul 19 16:47:20.660	INFO	xkflynhvm6hfx2	Disk space check passed: 79.98GB free (100.0% available)
Jul 19 16:47:20.660	INFO	xkflynhvm6hfx2	Network connectivity passed: Connected to 8.8.8.8 (44ms)
Jul 19 16:47:20.660	INFO	xkflynhvm6hfx2	CUDA version check passed: 12.8 (minimum: 11.8)
Jul 19 16:47:20.660	INFO	xkflynhvm6hfx2	CUDA initialization passed: 2 device(s) initialized successfully
Jul 19 16:47:20.660	INFO	xkflynhvm6hfx2	GPU compute benchmark passed: Matrix multiply completed in 96ms
Jul 19 16:47:20.660	INFO	xkflynhvm6hfx2	All fitness checks passed. (3644.25ms)
Jul 19 16:47:20.660	INFO	xkflynhvm6hfx2	Jobs in queue: 1
Jul 19 16:47:20.899	INFO	xkflynhvm6hfx2	Jobs in progress: 1
Jul 19 16:47:20.900	INFO	xkflynhvm6hfx2	Started.
Jul 19 16:47:20.900	INFO	xkflynhvm6hfx2	[v3-sequential-offload] Generating 512x512: A futuristic city with flying cars at night, neon reflections, cyberpunk
Jul 19 16:47:20.900	INFO	xkflynhvm6hfx2	The cache for model files in Transformers v4.22.0 has been updated. Migrating your old cache. This is a one-time only operation. You ca...
Jul 19 16:47:21.913	INFO	xkflynhvm6hfx2	\r0it [00:00, ?it/s]\r0it [00:00, ?it/s]
Jul 19 16:47:21.913	INFO	xkflynhvm6hfx2	[v3-sequential-offload] GPU: NVIDIA H200 (139.8 GB VRAM)
Jul 19 16:51:39.234	INFO	xkflynhvm6hfx2	\rFetching 23 files: 0% 0/23 [00:00<?, ?it/s]\rFetching 23 files: 9% 2/23 [00:00<00:05, 3.60it/s]\rFetching 23 files: 17% 4/23 [00:00<00:04, 5.40it/s]
Jul 19 16:51:39.426	INFO	xkflynhvm6hfx2	\rLoading pipeline components...: 0% 0/7 [00:00<?, ?it/s]
Jul 19 16:51:39.426	INFO	xkflynhvm6hfx2	\rLoading checkpoint shards: 0% 0/2 [00:00<?, ?it/s]\r
Jul 19 16:51:39.626	INFO	xkflynhvm6hfx2	\rLoading checkpoint shards: 50% 1/2 [00:00<00:00, 9.92it/s]\r\rLoading checkpoint shards: 100% 2/2 [00:00<00:00, 19.84it/s]
Jul 19 16:51:39.626	INFO	xkflynhvm6hfx2	\rLoading pipeline components...: 14% 1/7 [00:00<00:01, 4.19it/s]\rLoading pipeline components...: 43% 3/7 [00:00<00:00, 11.27it/s]
Jul 19 16:51:40.112	INFO	xkflynhvm6hfx2	\rLoading pipeline components...: 71% 5/7 [00:00<00:00, 9.25it/s]\rLoading pipeline components...: 100% 7/7 [00:00<00:00, 18.50it/s]
Jul 19 16:52:39.253	INFO	xkflynhvm6hfx2	[v3-sequential-offload] Model ready. GPU allocated: 0.03 GB (should be <2 GB)
Jul 19 16:52:39.254	INFO	xkflynhvm6hfx2	\r 0% 0/20 [00:00<?, ?it/s]\r 5% 1/20 [00:03<01:03, 3.36s/it]\r 10% 2/20 [00:06<00:54, 3.01s/it]\r 15% 3/20 [00:08<00:48, 2.67s/it]
Jul 19 16:52:40.131	INFO	xkflynhvm6hfx2	Finished

Figure: Logs - model load + 512x512 cyberpunk generation (v3-sequential-offload)

7.6 Metrics

- 5 completed requests, 0 failed
- ~8 min total execution time (includes cold model loads)
- 6 cold starts, ~56 s total cold-start time

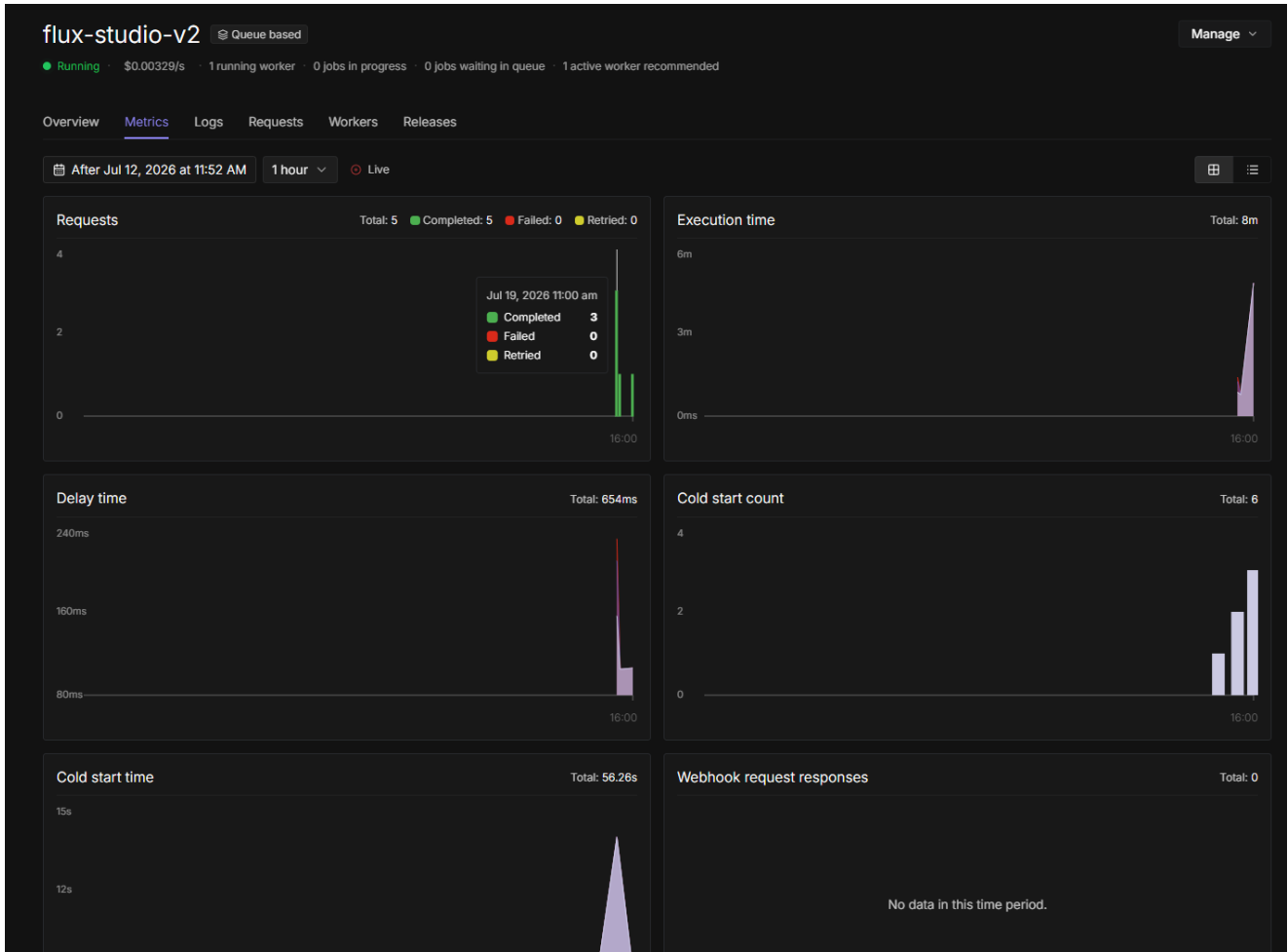


Figure: Metrics - 5 completed requests, execution time, cold starts

7.7 Release history

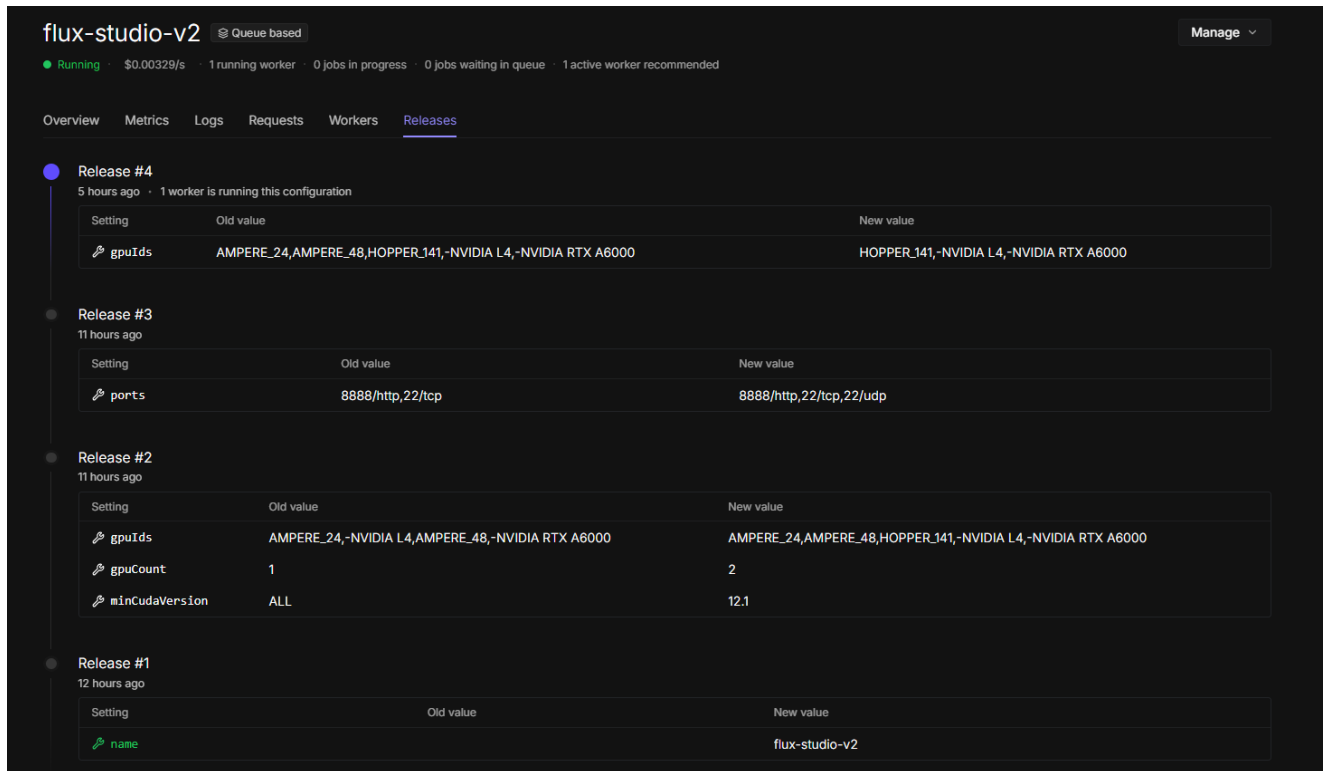


Figure: Release history - GPU filter updates, CUDA 12.1, worker scaling

8. Phase 4 - Chat Application

Backend (app/backend/main.py)

- GET /api/health - verify RunPod configuration
- POST /api/chat/submit - submit async job, return job_id
- GET /api/chat/status/{job_id} - poll until COMPLETED

Frontend features

- New chat, recent history, pin chats, projects/folders
- Library of all generated images
- Search chats (Ctrl+K)
- Settings: steps, guidance, resolution, seed
- Download PNG and copy prompt per image

8.1 Working UI

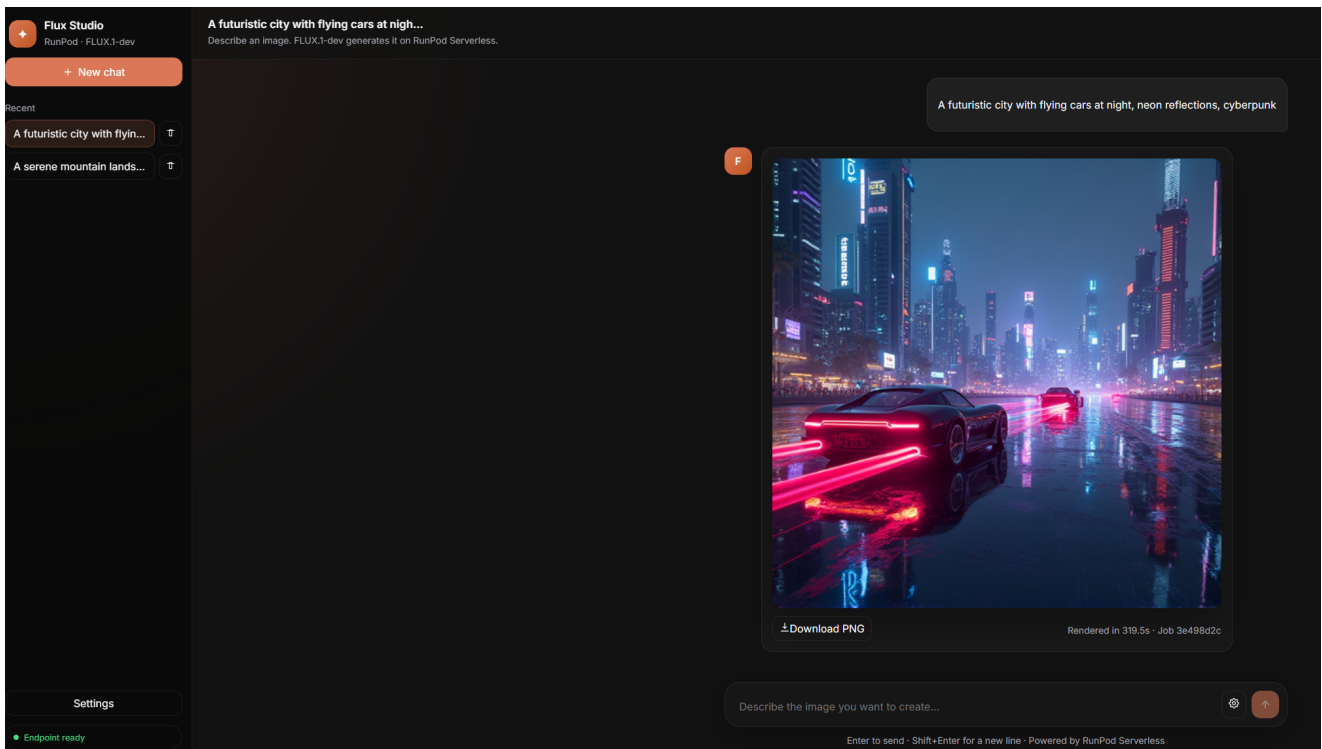


Figure: Flux Studio chat UI - successful image generation (319.5s render)

9. Phase 5 - Web Deployment (Render + Vercel)

Production: Vercel (frontend) -> Render (backend) -> RunPod (GPU)

Render backend

- Root Directory: app/backend
- Language: Python 3 (not Docker)
- Build: pip install -r requirements.txt
- Start: uvicorn main:app --host 0.0.0.0 --port \$PORT
- Env: RUNPOD_API_KEY, RUNPOD_ENDPOINT_ID, FRONTEND_URL

Vercel frontend

- Root Directory: app/frontend
- Build: npm run build - Output: dist
- Env: VITE_API_BASE_URL=https://your-api.onrender.com

Important: Do not use 'cd app/frontend && npm run build' when Root Directory is already app/frontend - that causes a Vercel build failure.

10. API Contract

RunPod input

```
{
  "input": {
    "prompt": "A futuristic city with flying cars at night",
    "width": 512,
    "height": 512,
    "num_inference_steps": 20,
    "guidance_scale": 3.5
  }
}
```

Handler output

```
{
  "image_base64": "<base64 PNG>",
  "prompt": "...",
  "width": 512,
  "height": 512,
  "handler_version": "v3-sequential-offload"
}
```

11. Challenges and Solutions

Problem	Fix	
HF gated model (403)	Accept FLUX license; set HF_TOKEN	
transformers 5.x breaks FluxPipeline	Pin diffusers 0.32.2, transformers 4.47.1	
CUDA driver mismatch	Pin torch cu124; minCudaVersion 12.1 on en	
CUDA OOM on 24 GB GPUs	Sequential CPU offload, 512 cap, VAE tilin	
Workers stuck initializing	80 GB container disk; correct Docker image	
Vercel build failure	Root Directory = app/frontend; build = npm	
Cold start latency	Set workersMin=1 on RunPod endpoint	

12. Performance Snapshot

- Cold start: ~56 s total (6 events observed)
- Full generation (cold, 512x512): ~319 s (UI screenshot)
- Warm worker (512x512): ~30-60 s
- Queue delay: ~100 ms
- Cost (141 GB tier): ~\$0.00329/s while worker runs

13. Submission Checklist

- handler.py implements prompt to base64 PNG
- Dockerfile builds and pushes to Docker Hub
- RunPod endpoint flux-studio-v2 running
- Logs show successful 512x512 generation
- Chat UI generates and displays images
- GitHub repo public
- Render backend deployed
- Vercel frontend deployed

GitHub: <https://github.com/germainssafari/Runpod>

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